

Question-1

Solve the following problem with cpu scheduling Algorithms.

Draw a table with 4 rows & with 3 columns having the following details.

Process	AT	BT
P ₁	0	5
P ₂	0	4
P ₃	0	3

There is a context switch in the cpu the context switch time is 5 seconds. Consider the context switch time B/w processes of 5 seconds.

Apply the following cpu scheduling Algorithms:-

* preemptive Algorithms:-

(i) SRTF

(ii) priority scheduling

* non-preemptive Algorithms:-

(i) fcf's

(ii) STP

(iii) priority scheduling

Solve each algorithm with its gantt chart, CT, TAT, WT, average TAT and Average WT.

Given,

process	AT	BT
P ₁	0	5
P ₂	0	4
P ₃	0	3

→ Context switch Time = 5 sec

formulas:-

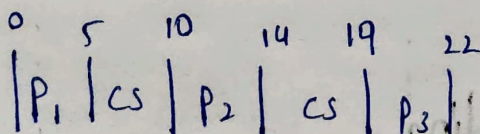
$$TAT = CT - AT$$

$$WT = TAT - BT$$

(i) fcf's (first come first serve)

order:- P₁ → P₂ → P₃

Gantt chart:-



Calculation:-

process	AT	BT	CT	TAT = CT - AT	WT = TAT - BT
P ₁	0	5	5	5	0
P ₂	0	4	14	14	10
P ₃	0	3	22	22	19

$$\text{Average WT} = \frac{0+10+19}{3} = 9.67$$

$$\text{Average TAT} = \frac{5+14+22}{3} = 13.67$$

(ii) SJF (shortest Job first)

Since SJF selects the process with the shortest

BT:-

P₃(3) → P₂(4) → P₁(5)

order :- P₃ → P₂ → P₁

Gantt chart:-

0 3 8 12 17 22
 | P₃ | C₃ | P₂ | C₂ | P₁ |

Calculation:-

process	AT	BT	CT	TAT	WT
P ₁	0	5	22	22	17
P ₂	0	4	12	12	8
P ₃	0	3	3	3	0

$$\text{Average WT} = \frac{17+8+0}{3} = 8.33$$

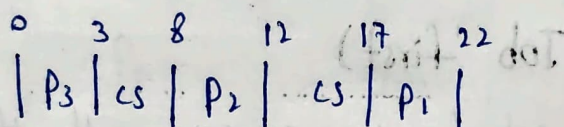
$$\text{Average TAT} = \frac{22+12+3}{3} = 12.33$$

(iii) SRTF

All process arrive at 0, so no preemption occurs

order: $P_3 \rightarrow P_2 \rightarrow P_1$

Gantt chart:



Calculation:-

process	AT	BT	CT	TAT	WT
P ₁	0	5	22	22	17
P ₂	0	4	12	12	8
P ₃	0	3	3	3	0

$$\text{Average WT} = 8.33$$

$$\text{Average TAT} = 12.33$$

(iv) priority scheduling

cannot solve because priority values are not given.

Question-2.

A mobile operating system has the following tasks

Task	AT	cpu Burst
Music Player(Mp)	0	20
Touch event(TF)	3	1
whatsapp notification (WN)	5	2
Camera launch (CL)	6	4

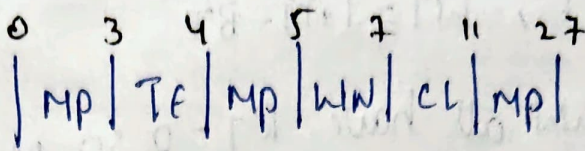
Questions:-

- i) Which scheduling algorithm is most suitable; FCFS, SJF, SRTF (or) RR?
- ii) Draw the execution timeline for SRTF.
- iii) Explain why the selected algorithm improves user experience.
- iv) Which task receives the fastest response time?
- (i) SRTF is the suitable scheduling algorithm because SRTF always selects the task with the shortest remaining cpu burst time.

(ii) SRTF Execution Timeline

- * In SRTF, the process with the shortest remaining burst time is selected.
- * 0-3: MP runs
Remaining MP = $20 - 3 = 17$
- * At 3: TE arrives with BT = 1
TE is shorter than MP $\rightarrow$ TE runs from 3-4
- * 4-5: MP continues
Remaining MP = 16.
- * At 5: WN arrives with BT = 2
WN is shorter than MP $\rightarrow$ WN runs
- * 6: CL arrives with BT = 4, but WN has only 1 unit remaining, so WN continues.
- * 5-7: WN completes
- * 7-11: Compare MP = 16 and CL = 4 $\rightarrow$ CL runs
- * 11-27: CL finishes, so remaining MP runs until completion.

→ Gantt chart



→ Final execution order

MP → TE → MP → WN → CL → MP

(iii) SRTF gives quick response to short tasks such as touch events, notifications and camera launch

Therefore, the mobile device feels:

- * Fast
- * Responsive
- * Interactive

→ SRTF improves responsiveness by prioritizing shorter remaining tasks

(iv) Response time

→ Response time = first cpu start time - AT

Task	AT	BT	first cpu start	CT	TAT	WT	RT
MP	0	20	0	27	27	7	0
TE	3	1	3	4	1	0	0
WN	5	2	5	7	2	0	0
CL	6	4	7	11	5	1	1

Where,

$$TAT = CT - AT ; WT = TAT - BT$$

→ MP, TE and WN all have $RT = 0$ so, Music player, touch event and Whatsapp notification share the fastest response time of 0 seconds.

Question 3:-

Define the following terms:-

(i) starvation:-

starvation occurs when a process waits for a very long time bcoz other processes are repeatedly selected for execution

eg:- A low priority process keeps waiting while high priority process continuously execute.

(ii) Convoy effect:-

Convoy effect occurs in FCFS scheduling when a long process makes many short process wait behind it

eg:- A long process runs first, causing several short processes to wait.

(iii) Context switching:-

It is the process of saving the state of the currently running process and loading the state of another process so that the CPU can switch between them.

Eg:- The OS saves process P_1 's state and loads P_2 's state.

(iv) Aging:-

Aging is a technique used to prevent starvation by gradually increasing the priority of the process that has been waiting for a long time.

Eg:- A low-priority process gets a higher priority as its waiting time increases.

Question-4:-

Consider the following process:-

Process	AT	BT	priority
P ₁	0	4	3
P ₂	0	9	1
P ₃	0	6	2
P ₄	0	4	3
P ₅	0	9	1
P ₆	0	6	2
P ₇	1	4	3
P ₈	1	9	1
P ₉	2	6	2

Apply the CPU scheduling algorithms to given processes

(i) FCFS

FCFS means first come first serve

Gantt chart

0 4 13 19 23 32 38 42 51 57
|P₁|P₂|P₃|P₄|P₅|P₆|P₇|P₈|P₉|

Table

process	AT	BT	CT	TAT	WT	RT
P ₁	0	4	4	4	0	0
P ₂	0	9	13	13	4	4
P ₃	0	6	19	19	13	13
P ₄	0	4	23	23	19	19
P ₅	0	9	32	32	23	23
P ₆	0	6	38	38	32	32
P ₇	1	4	42	41	37	37
P ₈	1	9	51	50	41	41
P ₉	2	6	57	55	49	49

Average:-

Average TAT = 30.56

Average WT = 24.22

ii) STF

STF means shortest job first. Since it is non-preemptive, once a process starts, it finishes completely.

Order:-

Shortest available processors are selected

$P_1 \rightarrow P_4 \rightarrow P_7 \rightarrow P_3 \rightarrow P_6 \rightarrow P_9 \rightarrow P_2 \rightarrow P_5 \rightarrow P_3$

Gantt chart

0 4 8 12 18 24 30 39 48 57
| P₁ | P₄ | P₇ | P₃ | P₆ | P₉ | P₂ | P₅ | P₃ |

Table

process	AT	BT	CT	TAT	WT	RT
P ₁	0	4	4	4	0	0
P ₄	0	4	8	8	4	4
P ₇	1	4	12	11	7	7
P ₃	0	6	18	18	12	12
P ₆	0	6	24	24	18	18
P ₉	2	6	30	28	22	22
P ₂	0	9	39	39	30	30
P ₅	0	9	48	48	39	39
P ₃	1	9	57	56	47	47

Average TAT = 26.22

Average WT = 19.89

(iii) SRTF

→ SRTF means shortest Remaining time first

→ It is the preemptive version of SJF

Execution

* 0-4 → P₁

* 4-8 → P₄

* 8-12 → P₇

* 12-18 → P₃

* 18-24 → P₆

* 24-30 → P₉

* 30-39 → P₂

* 39-48 → P₅

* 48-52 → P₈

Gantt chart

0 4 8 12 18 24 30 39 48 52

|P₁|P₄|P₇|P₃|P₆|P₉|P₂|P₅|P₈|

process	CT	TAT	WT	RT
P ₁	4	4	0	0
P ₄	8	8	4	4
P ₇	12	11	7	7
P ₃	18	18	12	12
P ₆	24	24	18	18
P ₉	30	28	22	22
P ₂	39	39	30	30
P ₅	48	48	39	39
P ₈	57	56	47	47

Average TAT = 26.22

Average WT = 19.89

(iv) priority scheduling

priority 1 = highest priority (ready time)

At time 0, P₂ and P₅ have the highest priority, so they are selected first.

→ Average TAT = 36.22

→ Average WT = 29.89

Order:-

$P_2 \rightarrow P_5 \rightarrow P_8 \rightarrow P_3 \rightarrow P_6 \rightarrow P_9 \rightarrow P_1 \rightarrow P_4 \rightarrow P_7$

Gantt chart:-

0 9 18 27 33 39 45 49 53 57
| P₂ | P₅ | P₈ | P₃ | P₆ | P₉ | P₁ | P₄ | P₇ |

Process	AT	BT	CT	TAT	WT	RT
P ₂	0	9	9	9	0	0
P ₅	0	9	12	18	9	9
P ₈	1	9	27	26	17	17
P ₃	0	6	33	33	27	27
P ₆	0	6	39	39	33	33
P ₉	2	6	45	43	37	37
P ₁	0	4	49	49	45	45
P ₄	0	4	53	53	49	49
P ₇	1	4	57	56	52	52